Project Final Summary: COVID-19 Impact on Unemployment

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Objective

Analyse unemployment data (May 2019 – June 2020) to measure COVID-19 impact, detect seasonal patterns, and inform policy.

Key Findings

1. **COVID-19 increased unemployment by 87%** (from 9.5% to 17.8% average).
   Policy: Income support and job creation programmes during health crises.

2. **April has the highest unemployment** (seasonal peak), exacerbated by COVID.
   Policy: Strengthen unemployment benefits before April each year.

3. **Rural unemployment rose more than urban** during COVID (rural +7.0 percentage points after accounting for month).
   Policy: Target rural employment schemes (e.g., MGNREGA).

4. **Highest COVID unemployment states**: Puducherry (39%), Jharkhand (36%), Haryana (35%), Bihar (32%), Tripura (27%).
   Policy: Regional relief prioritisation.

5. **Model performance** (R^2 = 0.28) – simple features explain 28% of variance. More data needed for better predictions.

Data Cleaning Summary

- Removed 28 empty rows and 27 duplicates.
- Standardised region names, converted dates.
- Final clean dataset: 740 rows, 6 columns.

Deliverables

- Cleaned dataset (CSV)
- Python notebook with all stages
- Visualisations (trend, monthly, rural/urban, residuals)
- Predictive model (Decision Tree) and interpretable linear regression coefficients
- Policy recommendations (above)

Next Steps for Intern

- Collect more years of data to isolate true seasonality.
- Add economic indicators (e.g., state GDP, lockdown stringency index).
- Deploy prediction API for real-time unemployment forecasting.

In [ ]:

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